NISHI MUDLIAR

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SUMMARY

5 years of experience in **Artificial Intelligence, Machine Learning, Data Analytics**, and Business Intelligence. Expertise in developing AI-powered applications, anomaly detection models, automation solutions, and data visualization dashboards. Strong proficiency in **Python, SQL, Power BI, and FastAPI.** Hands-on experience with **LLMs, feature selection, PCA, LangChain, AI agents** and cloud technologies.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, FastAPI, SQL, Power BI, Pandas, NumPy, Scikit-learn, PyTorch, LangChain, LangGraph, Hugging Face Transformers, Streamlit
- **Machine Learning & AI:** Supervised & Unsupervised Learning, Meta-learning, PCA, Feature Engineering, Anomaly Detection, Isolation Forest, Predictive Modeling, Classification, Regression, Clustering, Deep Learning, LLMs (GPT-4), Retrieval-Augmented Generation (RAG), Prompt Engineering, NLP
- **Data Processing & Visualization:** Exploratory Data Analysis (EDA), Power BI Dashboards, DAX, Excel Macros, Pivot Tables, Data Wrangling, Data Cleansing
- **Databases & Cloud:** Cosmos DB, SQL Server, AWS (S3, SageMaker, Lambda, Step Functions), Microsoft Azure (Cognitive Services, Azure Functions)
- **Automation & Deployment:** Git, CI/CD, MLflow, Docker, FastAPI, LangChain Agents, BeautifulSoup, API Development, Web Scraping

PROFESSIONAL EXPERIENCE

Data Scientist (29st September 2025 – Present)
[Barclays] -

Financial Crime Screening

- Working in financial payments and customer-sanctions screening, using python automation techniques to detect fraudulent or suspicious transactions in high-volume payment workflows.
- Performing customer sanctions screening by cross-checking entities against global watchlists, sanctions lists, and regulatory datasets, ensuring strong compliance with AML.

AI ML Developer (1st April 2024 – 26th September 2025)
[Siemens Energy Industrial Turbomachinery India Private Limited]

AI-Powered Document Query System

- Developed a Python-based document intelligence platform using **GPT-4**, enabling extraction of actionable insights from industrial documents.
- Integrated NLP-powered semantic search, boosting document accessibility by 40%.
- Designed and implemented automated rule generation pipelines using LLMs and pattern-mining techniques, reducing manual effort in document processing and enabling scalable knowledge extraction.
- **Fine-tuned OpenAI's LLM** with domain-specific corpora to enhance accuracy for technical queries.
- Built a Retrieval-Augmented Generation (**RAG**) system with vector embeddings, delivering precise, real-time answers with contextual grounding.

- Developed schema inference models to dynamically identify and adapt underlying data structures in heterogeneous industrial documents, improving downstream query accuracy and system interoperability.
- Engineered **LangGraph**-based **AI agents** to autonomously perform web search, scrape top-ranking pages across multiple domains, and generate summarized **insights**, enabling dynamic real-time knowledge augmentation.

Anomaly Detection in Compressor Systems

- Designed ML models to **detect anomalies** in dry gas seal behavior using 30–40 engineered features from temperature and pressure data.
- Employed **Isolation Forest, meta-learning ensembles, and PCA** to increase model performance across multiple equipment types.
- Accelerated model training by 40% and cut analysis time through **parallel processing** and feature optimization.
- Deployed a real-time **anomaly detection** pipeline on **AWS** (SageMaker, Step Functions, S3) to monitor 40+ compressor signals for **predictive maintenance**.

Data Analyst | Accenture – (21st Sep 2020 – 29th March 2024)

Machine Learning & Data Intelligence

- Built predictive models on 50+ raw datasets using Python (scikit-learn, Pandas) to uncover patterns and generate business insights.
- Automated large-scale data pipelines (800MB+) with SQL and Python, streamlining end-to-end ML workflows and enabling faster experimentation.
- Engineered robust feature extraction and transformation for 60,000+ customer records, improving personalization and boosting retention by 70%.

AI & Advanced Analytics

- Developed 25+ interactive dashboards in Power BI, embedding ML-based forecasting and KPI tracking for executive decision-making.
- Unified structured and semi-structured data from SQL, Excel, and APIs, preparing it for machine learning models and BI visualizations.
- Integrated OpenAI APIs for automated report summarization and narrative insights, enhancing stakeholder communication and saving analyst hours.

EDUCATION

Bachelor of Engineering – Computer Science & Engineering
CGPA: 9.28 | 06/2016 – 06/2020

CERTIFICATION

- **Microsoft Certified: Data Analyst Associate (DA-100)**
- **Microsoft Azure Fundamentals (AZ-900)**
- **Microsoft Power Platform Fundamentals (PL-900)**
- **Microsoft Azure Administrator (AZ-104)**
- **NPTEL – Python for Data Science (IIT Madras)**
- **Data Science with AI – UpGrad**